

Amudi Purba

amudipurba12@gmail.com | [+62-821-1557-5545](tel:+62-821-1557-5545) | [LinkedIn](#) | [GitHub](#)

Third-year Information Systems & Technology student at Institut Teknologi Bandung with a focused interest in data and artificial intelligence. Possesses a well-rounded foundation built through academic coursework in technical domains and organizational experience that has sharpened collaboration, leadership, and accountability skills. Understands the full lifecycle of building a product from the ground up, from ideation to execution, and applies that systems thinking to approach data-driven problems in a structured way. Known as a fast learner who adapts quickly to new environments and tools, with a genuine curiosity to explore and understand how things work. Eager to contribute to a team where technical depth and interpersonal skills are equally valued.

EDUCATION LEVEL

Institut Teknologi Bandung — Bandung, Indonesia

Aug 2023 – Aug 2027 (Expected)

Undergraduate Student of Information System and Technology

CGPA: 3.64/4.00 (128 credits)

- Relevant Courseworks: Database Modelling, Database Management, Foundation of Artificial Intelligence, Artificial Intelligence for Business, Web and Mobile Application Development, Enterprise Requirements Analysis, Enterprise Analysis.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C, PHP, Haskell

Frontend: React, React Native, NextJS, Expo, HTML, CSS, Tailwind CSS, JavaFX

Backend: FastAPI, Node.js, Express.js, RESTful API, JWT, Supabase

Database: PostgreSQL, MySQL — schema design, query optimization, relational modelling

DevOps & Tools: Git, GitHub, Docker, Postman, Swagger, Cloudflare, version control workflows

AI/ML Integration: YOLOv8, scikit-learn, LLM API integration, AI-as-a-Service

PROFESSIONAL EXPERIENCE

Quality Assurance, PT Akhdani Reka Solusi — Bandung, Indonesia

June 2026 – Aug 2026

- Identified and documented 40+ software defects across the testing lifecycle, reducing production bug risk and contributing to a stable pre-release product.
- Validated that features met business requirements and quality standards before release, supporting delivery of a user-friendly application on schedule.

Computational Thinking Lab Assistant, ITB

Sept 2025 – Dec 2025

- Evaluated Python programming assignments and provided feedback on computational thinking concepts.
- Evaluate the code of practical results and supervise practical experimental demonstrations.

Data Migration, ITB Education Directorate

July 2025 – Aug 2025

- Evaluated Python programming assignments and provided feedback on computational thinking concepts.
- Evaluate the code of practical results and supervise practical experimental demonstrations.

Basic Physics Lab Assistant, ITB 2024, LFD ITB

Sept 2024 – Nov 2024

- Improved report quality and student comprehension through structured evaluation and feedback mechanisms.
- Enhanced laboratory learning effectiveness by guiding students through experimental procedures and scientific reasoning.

PROJECT

Freelance: KJPP Consultant Ecosystem Application

2026

Developed a centralized internal ecosystem application for KJPP Hari Utomo aimed at improving operational coordination and reducing dependency on fragmented communication platforms such as WhatsApp, improving process centralization and organizational efficiency through more structured digital operations.

- Built frontend interfaces using TypeScript and React within a 3-member development team, supporting integrated document submission, file management, and attendance workflows in a unified platform.

[AI-Assisted CV Screening Application](#)

2026

Developed an AI-powered candidate classification system using multi-model AI integration (AI-as-a-Service + LLM components) within a 5-member cross-functional team.

- Designed data preprocessing and feature engineering pipelines for structured CV data, improving model input quality.
- Evaluated Logistic Regression model performance using precision, recall, and F1-score metrics
- Integrated LLM-based components to enable semantic understanding of unstructured resume text.

Parking Slot Detection using Computer Vision 2026

Built a real-time parking availability monitoring system using AI object detection and server-side communication in a 5-member team.

- Engineered a Python/FastAPI backend integrating YOLOv8-based detection
- Optimized server response pipeline to maintain low-latency data delivery under continuous detection load.

Cross-Platform Barbershop Reservation Application 2025

Developed a cross-platform reservation application for Brocode Barbershop aimed at improving queue management and customer booking convenience.

- Worked as Frontend Developer within a 2-member team, building responsive web and mobile interfaces using TypeScript, JavaScript, Expo, and Supabase with real-time booking functionality.
- Improved customer accessibility by enabling remote reservation scheduling, reducing the need for on-site waiting and enhancing overall service usability.

Fraud Analysis using AI Modelling 2025

Academic case study applying supervised ML models to detect fraudulent patterns from structured financial datasets.

- Built end-to-end ML pipeline: data cleaning, dataset splitting, feature preprocessing, model training (DTL, KNN, Logistic Regression), and evaluation using accuracy, precision, and recall.
- Compared model performance across algorithms to identify the best-fit classifier for imbalanced fraud data.

Enterprise Architecture Analysis 2025

Conducted enterprise architecture analysis for “Toko Kopi Manusia” using the TOGAF framework to evaluate operational readiness across business, data, application, and technology domains.

- Provided strategic technology recommendations including POS systems, Power BI integration, and technology management concepts to improve operational efficiency and business responsiveness.

Whoosh Database Analysis 2024

Conducted database analysis on the Whoosh railway system to understand data structures, entity relationships, and query-oriented database design.

- Strengthened practical understanding of database modeling, relational data design, and structured data management workflows.

ORGANIZATIONAL EXPERIENCE

IT TECFest 2025, Techno Entrepreneurship Club May 2025 – Dec 2025

- Improved report quality and student comprehension through structured evaluation and feedback mechanisms.
- Enhanced laboratory learning effectiveness by guiding students through experimental procedures and scientific reasoning.

Board of Director, Techno Entrepreneurship Club Dec 2024 – Present

- Led governance structuring initiatives to ensure organizational sustainability and alignment with long-term strategic goals.
- Improved cross-division coordination efficiency, enabling smoother execution of club initiatives and member programs.

AWARD

[National] Top 15 Finalist of Hackathon by SIMKOPDES 2026

- Developed KopHabit, a digital transformation platform for Koperasi Desa Merah Putih driving grassroots community participation through OCR-based bookkeeping and accessible financial transparency.

[National] Bank Indonesia Scholarship Awardee 2026

CERTIFICATIONS

Foundation of Machine Learning— [\[Certificate\]](#) March 2026

Introduction to SQL — [\[Certificate\]](#) March 2026

Introduction of Data Science — [\[Certificate\]](#) March 2026

Foundation of Artificial Intelligence — [\[Certificate\]](#) Feb 2026

Start Code with Python — [\[Certificate\]](#) Feb 2026

Generative AI Certified by Microsoft— [\[Credential\]](#) July 2024